

Joseph Resch

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Education

University of California, Los Angeles (UCLA), Los Angeles, CA 2020 – 2026
Ph.D. in Statistics | GPA: 3.94 | University Teaching Fellow

University of Georgia, Athens, GA 2016 – 2020
B.A. Economics | B.S. Statistics | Undergraduate Research Fellow

Skills

Languages: Python, R, SQL, Bash, C++, SAS

ML & Data: Pandas, NumPy, scikit-learn, PyTorch, XGBoost, LightGBM, Optuna, SHAP, Rcpp, Shiny

Tools & Cloud: AWS (SageMaker, S3), Snowflake, Git, Tableau, JMP

Methods: Machine Learning, Bayesian Modeling, A/B Testing & Experimental Design, Causal Inference, Simulations, Hierarchical Models, Latent-Variable Models, MCMC, Variational Inference, Segmentation

Experience

Doctoral Researcher, Statistics & Machine Learning 2020 – 2026
UCLA Statistics & Data Science

- Developed scalable Bayesian inference for exponential-family random graph models (ERGMs), enabled estimation on social and economic networks too large for standard methods; published first-author in PLOS ONE.
- Derived a Bayesian reference prior and implemented Metropolis–Hastings MCMC and variational-Bayes estimators, improved goodness-of-fit ~15%; released software on GitHub: [BayesianNDGPriorERGM](#), [NondegeneracyPriorERGM](#).
- Built a simulation benchmarking framework for likelihood-approximation methods and optimized it for large-scale runs on HPC clusters; released software on GitHub: [VariationalEstimationERGM](#).

Data Science Intern 2025
Farmers Insurance

- Built XGBoost and LightGBM models for insurance pricing and risk segmentation that identified the top auto-risk decile with over 2× the lift of the production GLM baseline.
- Shipped the models to production, where they drive marketing experimentation and targeted high-risk campaigns.
- Engineered features from 80M+ row Snowflake SQL pipelines, built reusable workflows for risk modeling at scale.
- Trained models on AWS SageMaker with GPU acceleration, tuned hyperparameters with Optuna, and used SHAP to make risk drivers interpretable for non-technical stakeholders.

Research Data Scientist 2022 – 2025
UCLA School of Education & Information Studies

- Led an IES (U.S. Dept. of Education)-funded project end to end, from problem framing to a first-author publication in *Psychometrika*: estimated causal intervention effects on students' latent skill acquisition using a Bayesian time-series model, validated on real educational testing data.
- Senior member of a three-person data science team; seeded two follow-on projects taken on by later-hired data scientists and onboarded them with internal code documentation.
- Derived an efficient Gibbs sampler using Pólya-Gamma augmentation, cut runtime ~30% and made inference tractable at scale.
- Validated parameter recovery; released software on GitHub: [BayesianTDCM](#).

Research Assistant 2018 – 2020
University of Georgia, Department of Statistics

- Built sequential Gaussian-process surrogate models with space-filling experimental designs to calibrate an expensive hydrological black-box simulator, improved fit ~20% over competing methods; co-authored a paper in the *Journal of Statistical Theory and Practice*.
- Ran large-scale parallelized simulations on an HPC cluster in R.

Selected Publications

"Assessing Variational Likelihood Estimation in Network Formation Models." *PLOS ONE* (2026). First author.

"Bayesian Transition Diagnostic Classification Models with Pólya-Gamma Augmentation." *Psychometrika* (2025).

First author.

"Solving an Inverse Problem for Time-Series-Valued Computer Simulators via Multiple Contour Estimation." *Journal of Statistical Theory and Practice* (2023). Co-author.